27th IMEKO TC4 Conference 2026 with Special TC11 Technical Session

Prague, 16. – 18. September 2026



Measuring the Future

IMEKO TC4 International Conference 2026
with TC11 Technical Session

BOOK OF ABSTRACTS

27th IMEKO TC4 Conference 2026 with Special TC11 Technical Session

Prague, 16. – 18. September 2026

BOOK OF ABSTRACTS

September 16-18, 2026

Prague, Czechia

© 2026 CTU in Prague

All rights reserved. No part of this publication may be reproduced in any form, nor may it be stored in a retrieval system or transmitted in any form, without written permission from the copyright holders.

Welcome Message from the Organizing Committee

On behalf of the Organizing Committee, we extend a warm welcome to you for the 27th IMEKO TC4 Conference 2026 with Special TC11 Technical Session.

The conference commences with a Welcome Ceremony and Get-Together Reception on September 16th, 2026, held in the Penthouse on the 10th floor of the CIIRC CTU building. Perched above Prague with its own terrace and a view across the rooftops to Prague Castle, the Penthouse — part of the Czech Institute of Informatics, Robotics and Cybernetics — offers an informal and relaxed setting in which to reconnect with old friends and meet new colleagues before the technical programme begins.

The Scientific Conference itself takes place on September 17th and 18th at the Czech Technical University in Prague, Faculty of Electrical Engineering, on the university's Dejvice campus. On the evening of the first conference day, delegates are invited to the IMEKO TC4 Gala Dinner at Restaurace Mlýnec, a restaurant occupying a distinctive spot on Novotného lávka with a terrace sitting directly beneath Charles Bridge and a view across the Vltava river toward Prague Castle. Part of the Zátiší Group and Michelin Selected, Mlýnec pairs a contemporary, water-inspired interior with a kitchen that blends traditional Czech technique and seasonal, local ingredients — a fitting way to celebrate a conference themed "Measuring the Future" in one of Europe's oldest university cities.

This edition also features a Special TC11 Technical Session dedicated to "Testing, Inspection and Certification in Electrical Sector," bringing the digital transformation of metrology, blockchain-based traceability and secure data infrastructures alongside TC4's traditional programme.

The main objective of the IMEKO TC4 conference is to foster collaboration between academic researchers and industry engineers, with a specific focus on measurement and instrumentation for electrical and magnetic quantities, ADC and DAC modelling, characterization and testing, calibration, traceability and metrological standards, advanced sensors and signal processing, and applications in energy systems, e-mobility, industry automation and emerging technologies.

IMEKO TC4's primary interest lies in the theoretical and practical aspects of measuring electrical quantities through electronic instruments, encompassing measurement accuracy, instrumentation design, calibration principles, uncertainty analysis and digital metrology.

We would like to take this opportunity to express our gratitude to all the members of the Technical Programme Committee, the Special Session Organizers, and the anonymous Reviewers, whose contributions have made this outstanding programme possible. We also extend our appreciation to the Authors for submitting their contributions and sharing their scientific findings with the IMEKO TC4 & TC11 community. Over the course of three days, the technical programme features six invited keynote lectures and about forty oral contributions from authors around the world, organized into eight thematic sessions.

Special thanks are due to all the local Organizing Committee members, whose significant efforts were devoted to the successful organization of this event, and to our sponsors and partners — Teste, HTEST, and Termovize TMV SS — for their generous support, which made this event possible.

The Organizing Committee:

General Conference Chair – Assoc. Prof. Jakub Svatos, Ph.D., CTU in Prague, Czechia
Scientific Chairs – Dr. Dragana Popovic Renella, SENIS AG, Switzerland;
Prof. Luca Callegaro, INRIM, Italy;
Prof. Carlos Roberto Hall Barbosa, PUC-Rio, Brazil

Technical Program Overview

The technical programme of the 27th IMEKO TC4 Conference 2026 with Special TC11 Technical Session evolves over three days (September 16-18, 2026) and includes six keynote speeches and eight oral sessions, including a Special TC11 Technical Session on "Testing, Inspection and Certification in Electrical Sector."

The keynote speeches are:

- One Bit to Useful Knowledge: Rethinking Data Acquisition for Low-Power and High-Speed Measurement Systems by Prof. Paolo Carbone, University of Perugia, Italy
- Major European Research Projects in Electrical Metrology by Prof. Jiří Tesař, Director of the Czech Metrology Institute (CMI), Czechia
- Measuring the Performance of C-UAS Systems: How Hard Can It Be? by Dr. Cristian Coman, NATO Communications and Information Agency (NCIA), The Hague, Netherlands
- Artificially indeterminate: measuring quantity, cost and impact of Artificial Intelligence by Dr. Michael Street, NATO Communications and Information Agency (NCIA), The Hague, Netherlands
- Trust by Design in Digital Metrology: From Machine-Readable Documents to Privacy-Preserving Computation by Prof. Kruno Miličević, J. J. Strossmayer University of Osijek, Croatia
- EUROLAB: Empowering Digital and Sustainable Labs and other CABs by Laura Martin, EUROLAB

The eight technical sessions cover the whole spectrum of electrical and magnetic measurements, from calibration and traceability to advanced sensors and emerging applications, alongside the Special TC11 Technical Session.

The oral sessions titles are:

- Session 1: Calibration, Traceability & Metrological Standards I
- Session 2: Calibration, Traceability & Metrological Standards II
- Session 3: Uncertainty Evaluation & Electrical/Magnetic Instrumentation
- Session 4: Electrical & Magnetic Instrumentation
- Session 5: ADC/DAC, Waveform & Data Conversion Metrology
- Session 6: Advanced Sensors & Signal Processing I
- Session 7: Advanced Sensors & Measurement Systems II
- Session 8 - Special TC11 Technical Session: Applications in Energy Systems, Emerging Technologies & Materials Characterization (incl. Testing, Inspection and Certification in Electrical Sector)

As Technical Program Chairs, we would like to thank all the members of the Technical Program Committee and the Reviewers, who allowed to improve the level of the contributions.

The Committee

IMEKO TC4 CHAIR

Assoc. Prof. Jakub Svatos, Ph.D., CTU in Prague, Czechia

GENERAL CONFERENCE CHAIR

Assoc. Prof. Jakub Svatos, Ph.D., CTU in Prague, Czechia

SCIENTIFIC CHAIRS

Dr. Dragana Popovic Renella, SENIS AG, Switzerland

Prof. Luca Callegaro, Istituto Nazionale di Ricerca Metrologica (INRIM), Italy

Prof. Carlos Roberto Hall Barbosa, Pontificia Universidade Catolica do Rio de Janeiro, Brazil

TECHNICAL PROGRAMME CO-CHAIRS

Prof. Jan Holub, Ph.D., CTU in Prague, Czechia

Dr. Jan Sobotka, Ph.D., CTU in Prague, Czechia

SPECIAL SESSIONS CHAIRS

Prof. Alvaro Silva Ribeiro, National Laboratory for Civil Engineering, Lisbon, Portugal

MSc. Mladen Jakovcic, Croatian Metrology Society, Croatia

Prof. Marija Cundeva-Blajer, Ss. Cyril and Methodius University in Skopje, North Macedonia

INTERNATIONAL PROGRAM COMMITTEE

Prof. Luca Callegaro, Istituto Nazionale di Ricerca Metrologica (INRIM), Italy

Prof. Marija Cundeva-Blajer, Ss. Cyril and Methodius University in Skopje, North Macedonia

Prof. Pasquale Daponte, University of Sannio, Italy

Prof. Cristian Fosalau, Technical University of Iasi, Romania

Prof. Vladimír Haasz, CTU in Prague, Czechia

Prof. Carlos Roberto Hall Barbosa, Pontificia Universidade Catolica do Rio de Janeiro, Brazil

Dr. Stanislav Karpenko, PTB, Germany

Prof. Raul Land, Tallinn University of Technology, Estonia

Prof. Theodore Laopoulos, Aristotle University of Thessaloniki, Greece

Dr. Dragana Popovic Renella, SENIS AG, Switzerland

Prof. Alexandru Salceanu, Technical University of Iasi, Romania

Prof. Ján Šaliga, Technical University of Košice, Slovakia

Assoc. Prof. Jakub Svatos, Ph.D., CTU in Prague, Czechia

About this Book of Abstracts

This Book of Abstracts contains the one-page extended abstracts presented at the 27th IMEKO TC4 Conference 2026 with Special TC11 Technical Session, held in Prague, Czechia, on September 16–18, 2026.

Contributions to the conference were submitted through the IMEKO conference management system (conferences.imeko.org) in two categories: full papers of 4–6 pages, which are published in the Proceedings of the conference (ISBN 978-92-990090-7-9), and one-page extended abstracts, which are collected in this volume. Of the 38 contributions presented at the conference, 28 were full papers and 10 were extended abstracts.

The abstracts are arranged in the order of their presentation in the technical programme and grouped by oral session. Each abstract is identified by its contribution number (# of paper), which is used in the conference management system and in the programme. Abstracts are marked with “o” in the programme, together with the page on which they appear.

The abstracts are reproduced as submitted by the authors, who are solely responsible for their content.

Program IMEKO-TC4 & TC11 Technical Session 2026

*o one-page extended abstract included in this Book of Abstracts [page] * full paper published in the Proceedings volume*

Day 1 - Wednesday, September 16, 2026		
17:00-19:00	Welcome Ceremony & Get-Together Reception	Penthouse - CIIRC, CTU
Day 2 - Thursday, September 17, 2026		
08:30-08:55	Conference Opening & Welcome Address	
Session 1: Calibration, Traceability & Metrological Standards I (Chair: Marija Cundeva-Blajer)		
08:55-09:30	**KEYNOTE — One Bit to Useful Knowledge: Rethinking Data Acquisition for Low-Power and High-Speed Measurement Systems**	*Prof. Paolo Carbone*
09:30-09:45	* Development of System for Disseminating AC/DC Voltage Difference Using Ranges of Transfer Standard and Investigating Stability	Valentyn Isaiev
09:45-10:00	* MetroMag: A European Infrastructure for Low Magnetic Field Metrology	Marco Coisson
10:00-10:15	o Franco Cabiati, electrical impedance metrology, and IMEKO [p. 1]	Luca Callegaro
10:15-10:30	* Exploring calibration schemes for electric impedance measurements beyond the short-open-load	Alessandro Cultrera
10:30-11:00	*Coffee Break*	
Session 2: Calibration, Traceability & Metrological Standards II (Chair: Luca Callegaro)		
11:00-11:30	**KEYNOTE — Major European Research Projects in Electrical Metrology**	*Prof. Jiri Tesar*
11:30-11:45	o High-Temperature Superconductor Josephson Junction Arrays Coupled to Microwave of Different Polarizations [p. 2]	Changhao Zhao
11:45-12:00	* MEMS-based Long-haul Optical Link Asymmetry Estimation for Atomic Clock Comparison	Michal Spacek
12:00-12:15	* A Model for Secure Exchange of Digital Calibration Certificates	Ahmed Khan Leghari
12:15-12:30	* Metrological Traceability in IoT Sensor Networks: A Microservice-based Framework for Automated Calibration Management	Marco Sento
12:30-12:45	* Implementation of Monte Carlo Simulation for measurement uncertainty estimation in electrical power and energy reference standards calibration	Kiril Demerdziev
12:45-13:00	TMV SS - Use of Thermography for Non-Destructive Testing	Antonin Krnoul
13:00-14:00	*Lunch Break*	

Session 3: Uncertainty Evaluation & Electrical/Magnetic Instrumentation (Chair: Paolo Carbone)		
14:00-14:30	**KEYNOTE — Measuring the Performance of C-UAS Systems: How Hard Can It Be?**	*Dr. Cristian Coman*
14:30-14:45	* GUM-compliant uncertainty evaluation in measurements based on ML regression or classification models	Leopoldo Angrisani
14:45-15:00	* Quality Infrastructure for Resilient Electrical Sector - Ensuring Confident Measurement Results in Calibration of High Currents Instruments	Marija Cundeva-Blajer
15:00-15:15	o Development of the Digital Twin for a Digital Impedance Ratio Bridge [p. 3]	Ana Lucia Medina Rodriguez
15:15-15:30	* Accuracy Analysis of THD Measurement of Power Signals by Means of a Low-cost Oscilloscope	Gabriel Gasparesc
15:30-16:00	*Coffee Break*	
Session 4: Electrical & Magnetic Instrumentation (Chair: Cesare Svelto)		
16:00-16:15	o The primary Tesla Standard at PTB [p. 4]	Franziska Weickert
16:15-16:30	o A traceable calibration of low-field magnetic coil systems in noisy industrial environments [p. 5]	Michal Ulvr
16:30-16:45	* Model-based magnetization measurement in Fluxgate sensors	Jiri Maier
16:45-17:00	* Estimation of torquer rod magnetic moment with fluxmetric method and compact sensing coils	Michal Janosek
17:00-17:15	* Compact modular cryogenic dielectric resonator for surface impedance measurements in high magnetic fields	Kostiantyn Torokhtii
19:00-21:00	Gala Dinner	

Day 3 - Friday, September 18, 2026

Session 5: ADC/DAC, Waveform & Data Conversion Metrology (Chair: Pedro M. Ramos)		
08:30-09:00	**KEYNOTE — Artificially indeterminate: measuring quantity, cost and impact of Artificial Intelligence**	*Dr. Michael Street*
09:00-09:15	* A systematic aliasing suppression architecture for real-time FI-DAC based on linear-phase complementary filter and oversampling	Yang Shen
09:15-09:30	* Parameter estimation of complex-valued sinewaves by a frequency-domain linearized sine-fit algorithm	Dominique Dallet
09:30-09:45	* Benchmarks for IIR Model Identification Based on Binary Measurements	Paolo Carbone
09:45-10:00	o A Digital Voltage Ramp Generator with Integrating Compensation [p. 6]	Jianting Zhao
10:00-10:30	*Coffee Break*	

Session 6: Advanced Sensors & Signal Processing I (Chair: Jan Holub)		
10:30-11:00	**KEYNOTE — Trust by Design in Digital Metrology: From Machine-Readable Documents to Privacy-Preserving Computation**	*Prof. Kruno Milicevic*
11:00-11:15	o Acoustic-Based Detection, Classification, and Localization of Impulsive Sounds [p. 7]	Martin Maxa
11:15-11:30	o Lock-in detection algorithm for signals modulated with irregular intervals [p. 8]	Shingo Tamaru
11:30-11:45	o A Wireless Acoustic Monitoring System for Drone Localization Based on MEMS Microphones and LoRa Communication [p. 9]	Kamil Herman
11:45-12:00	* Comparative Analysis of Search Algorithms for Automatic L-type Matching Network	Joao Pedro Torres S. Andrade
12:00-12:15	* Airborne Navigation During GNSS Outages Using Tightly Coupled IMU/DVB-T2 Fusion	Michal Koldinsky
12:15-13:15	*Lunch Break*	
Session 7: Advanced Sensors & Measurement Systems II (Chair: Luca De Vito)		
13:15-13:30	* Microcontroller Based Software Defined Instruments as an Economical Substitute of Virtual Measurement Instruments	Jakub Svatos
13:30-13:45	* Exploring a new indirect approach to latency measurement	Jakub Turinsky
13:45-14:00	* Detection of Mercury Ion in Water Samples Using a Low-Tech EPS-Coated Interdigitated Capacitive Sensor through S-Parameter Analysis	Henrique Dantas Silva
14:00-14:15	* Development and Experimental Evaluation of a Three-Lead ECG Holter System	Ondrej Kovac
14:15-14:30	* Multiparametric metrological characterization of an inductive sensing element for mechanical wear estimation	Davide Ceschini
14:30-15:00	*Coffee Break*	
Session 8: Applications in Energy Systems, Emerging Technologies & Materials Characterization (Chair: Gabriel Gasparesc)		
15:00-15:15	**KEYNOTE — EUROLAB: Empowering Digital and Sustainable Labs and other CABs**	*Laura Martin*
15:15-15:30	* Detection of Physical Layer Anomalies and Eavesdropping in Real World Data Cables	Amit Duwal
15:30-15:45	* Sustainability-oriented optimization of a sensing board for applications as a cyber-physical measurement system (CPMS)	Leopoldo Angrisani
15:45-16:00	* Vector Fitting-Based Feature Extraction from Electrochemical Impedance Spectroscopy for SOC and SOH Estimation	Pedro Ramos
16:00-16:15	* Measurement of the characteristics of EDL and hybrid supercapacitors versus temperature	Ornella Conti
16:15-16:30	o Non-Destructive Microwave Dielectric Characterization of Leather for Automotive Applications [p. 10]	Kostiantyn Torokhtii
16:30-16:45	* Measurements of Wetsuits Thermal Properties at 100 kPa and 600 kPa Absolute Pressure	Cesare Svelto
16:45-17:00	Closing Remarks & Best Paper Awards	

CONTENT

# of paper	Title and authors	Page number
Session 1: Calibration, Traceability & Metrological Standards I		
36	Franco Cabiati, electrical impedance metrology, and IMEKO *Luca Callegaro, Gian Carlo Bosco, Vincenzo D'Elia*	1
Session 2: Calibration, Traceability & Metrological Standards II		
78	High-Temperature Superconductor Josephson Junction Arrays Coupled to Microwave of Different Polarizations *Changhao Zhao, Da Xu, Yafang Lian, Yang Shi, Qing He, Jianting Zhao*	2
Session 3: Uncertainty Evaluation & Electrical/Magnetic Instrumentation		
43	Development of the Digital Twin for a Digital Impedance Ratio Bridge *A. L. Medina Rodriguez, Jan Kučera*	3
Session 4: Electrical & Magnetic Instrumentation		
57	The primary Tesla Standard at PTB *Franziska Weickert, Zhaowen Liu, Rainer Ketzler, Wiebke Pohlandt, Wolfgang Kilian*	4
64	A traceable calibration of low-field magnetic coil systems in noisy industrial environments *Michal Janosek, Michal Dressler, Michal Ulvr*	5
Session 5: ADC/DAC, Waveform & Data Conversion Metrology		
80	A Digital Voltage Ramp Generator with Integrating Compensation *Yaqiong Fu, Xianhui Li, Kenan Xu, Qing He, Jianting Zhao*	6
Session 6: Advanced Sensors & Signal Processing I		
28	Acoustic-Based Detection, Classification, and Localization of Impulsive Sounds *Jakub Svatos, Martin Maxa*	7
32	Lock-in detection algorithm for signals modulated with irregular intervals *Shingo Tamaru*	8
79	A Wireless Acoustic Monitoring System for Drone Localization Based on MEMS Microphones and LoRa Communication *Kamil Herman, Jakub Svatos, Martin Maxa*	9
Session 8: Applications in Energy Systems, Emerging Technologies & Materials Characterization		
55	Non-Destructive Microwave Dielectric Characterization of Leather for Automotive Applications *Andrea Alimenti, Kostiantyn Torokhtii, Alessandro Magalotti, Serena De Santis, Veronica Granata, Nicola Pompeo, Vincenzo Roberti, Daniele Rocco, Enrico Silva*	10

Franco Cabiati, electrical impedance metrology, and IMEKO

Luca Callegaro[1,2], Gian Carlo Bosco[1], Vincenzo D'Elia[1,2]

[1]*INRIM–Istituto Nazionale di Ricerca Metrologica, Torino, Italy. l.callegaro@inrim.it*
[2]*previously IEN–Istituto Elettrotecnico Nazionale "Galileo Ferraris"*

I. INTRODUCTION

Franco Cabiati (1938–2026) was an Italian electrical metrologist. He contributed to the development of primary metrology of electrical impedance, to the revision of the International System of Units and the determination of fundamental constants of nature, as well as to the topic of measurement uncertainty expression. Several of his results were presented at the IMEKO World Congresses and IMEKO TC-4 symposia. Here we briefly recall his contributions to electrical impedance metrology, with a focus on presentations at IMEKO events.

II. IMPEDANCE METROLOGY

The realisation of the impedance units and scales was a major focus of Cabiati's investigations. He worked on the determination of the uncertainty of the calculable capacitor, and subsequently on the metrology of the quantized Hall resistance in the AC regime at audio frequencies.

In primary impedance metrology, comparisons and scaling are traditionally performed with coaxial transformer impedance bridges, which are passive networks of electromagnetic components. Cabiati was a pioneer of the introduction of mixed-signal electronics in impedance comparison systems. He developed a fully-digital impedance bridge for LC comparisons based on a two-phase precision digital sinewave generator. A major contribution was the application of the *three-voltage method*, previously employed in electrical power measurement, to the comparison of impedances having arbitrary phase angles [1, 2]. To improve the accuracy of his impedance comparison setups, he developed the theory and application of multiple-winding transformers, suitable for use as current sources and for performing vector voltage operations [3]. He developed techniques for the definition of four-terminal impedances with active multi-coaxial cables [4], which reduce errors due to connections to negligible levels.

BIOGRAPHY

Franco Cabiati was born in Alessandria (Piemonte, Italy) in 1938. He graduated from the Polytechnic University of Turin in 1964. In 1965 he joined the Istituto Elettrotecnico Nazionale "Galileo Ferraris" (IEN; from 2006 Istituto Nazionale di Ricerca Metrologica, INRIM) as a researcher, working on electrical impedance and power measurement systems.

In 1972, he was a guest researcher at the U.S. National Bureau of Standards (NBS, now National Institute for Standards and Technology, NIST) on calculable capacitors and precision impedance comparisons. He later designed advanced metrology laboratories for maintaining national electrical standards, featuring strict thermal control, electromagnetic shielding, and vibration isolation.

He was appointed head of Electrical Metrology of IEN in 1988. He contributed to the development of international standards, notably influencing the adoption of the conventional voltage and resistance units in 1990. He proposed alternative methods to the watt balance experiment for determining the Planck constant, based on purely electrical forces and levitation concepts.

His research spanned AC electrical measurements, impedance comparison systems, and international metrology collaborations. In later years, he contributed to projects linking capacitance measurements to the quantum Hall effect and proposed advanced architectures for impedance standards. After retirement, he continued to provide significant theoretical contributions, including participation in CODATA's Task Group on Fundamental Constants.

Franco Cabiati passed away on 7 April 2026.

REFERENCES

[1] F. Cabiati, G. C. Bosco, and A. Sosso, "Impedance comparison in the low to medium range through precision ac voltage measurements," in *Proc. IMEKO XIII World Congress*, Torino, Italy, Sep 5-9 1994.

[2] F. Cabiati, L. Callegaro, A. Muciek, G. C. Bosco, and V. D'Elia, "Impedance measurement by an improved three-voltage method," in *Proceedings of the 9th IMEKO TC-4 Symposium on Electrical Measurements in Industry*, Glasgow, Scotland, Sep 8-9 1997.

[3] F. Cabiati, U. Pogliano, and G. C. Bosco, "Precision vector-difference transformer with high common-mode rejection," in *Proc. 11th IMEKO TC-4 Symposium on Trends in Electrical Measurements and Instrumentation*, Lisbon, Portugal, Sep 13-14 2001.

[4] F. Cabiati, V. D'Elia, and J. Bohaček, "Application of active compensated four-coaxial cables to AC resistance standards comparison," in *Proc. 12th IMEKO TC-4 International Symposium on Electrical Measurements and Instrumentation*, Zagreb, Croatia, Sep 25-27 2002.

High-Temperature Superconductor Josephson Junction Arrays Coupled to Microwave of Different Polarizations

Changhao Zhao[1], Da Xu[1], Yafang Lian[2], Yang Shi[1], Qing He[1], Jianting Zhao[1,†]

[1]*National Institute of Metrology, Beijing 100029, China*
[2]*China Jiliang University, Hangzhou 310018, China*
[†]*zhaojt@nim.ac.cn*

***Abstract* – YBCO bicrystal Josephson junction arrays were coupled to ∼ 70 GHz Fabry-Pérot resonator to synthesize Shapiro steps. Electromagnetic simulations and experimental tests showed that when the electric field is polarized parallel to the bicrystal grain boundary, coupling effect is promoted compared to the case of perpendicular polarization, which can be explained by dipole antenna coupling mechanism.**

Keywords: high-temperature superconductor, bicrystal Josephson junctions, Fabry-Pérot resonator, dipole antenna

I. INTRODUCTION

Shapiro steps up to 25–100 mV have already been synthesized with YBCO bicrystal Josephson junction arrays driven by Fabry-Pérot resonators [1,2]. However, the mechanism behind optimal polarization has not been well explained.

II. RESULTS AND DISCUSSIONS

Electromagnetic simulation result of an array of bicrystal junctions coupled to Fabry-Pérot resonator is shown in figure 1. When the electric component of the field was polarized parallel to the grain boundary, the average magnitude of microwave current induced in the junctions and the uniformity factor $s(f) = 1/\log_{10}[I_{\mathrm{rf}}^{\max}(f)/I_{\mathrm{rf}}^{\min}(f)]$, where $I_{\mathrm{rf}}^{\max}$ and $I_{\mathrm{rf}}^{\min}$ are the maximum and minimum microwave current induced in the junction array, were promoted compared to the case of perpendicular polarization. Measurement of YBCO bicrystal Josephson junctions driven by Fabry-Pérot resonator at 77 K confirmed the distinction, as shown in figure 2.

Dipole antenna scattering in the case of perpendicular polarization lead to the varied coupling strength, where I_{rf} at the center of the array are diminished due to scattering of the adjacent dipole antennas. With parallel polarization, however, the whole array works as a single dipole antenna thus coupling uniformity can be significantly improved. A clearer picture of the microwave coupling mechanism would facilitate junction array design in the future.

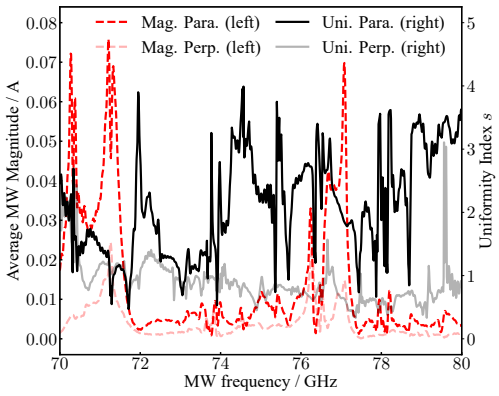


Fig. 1. Simulated coupling strength and uniformity.

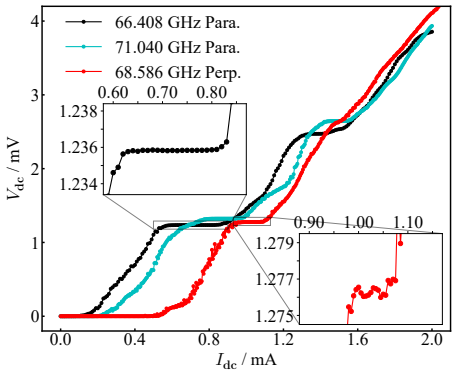


Fig. 2. Measured Shapiro steps with different polarization.

REFERENCES

[1] M. He, A. M. Klushin, S. L. Yan, N. Klein, "Optimization of Bicrystal Josephson Junctions and Arrays in a Fabry-Perot Resonator", IEEE Trans. on Appl. Supercond., vol.17, No.2, June 2007, pp.934-937.

[2] A. M. Klushin, M. He, M. Yu Levitchev, V. V. Kurin, N. Klein, "Optimization of the coupling of mm wave power to arrays of high-T_c Josephson junctions", Journal of Physic: Conference Series, vol.9, 2008, 012268.

Development of the Digital Twin for a Digital Impedance Ratio Bridge

A.L. Medina Rodriguez[1,2], Jan Kučera[1]

[1] *CMI - Czech Metrology Institute, Okruzni 31, 638 00 Brno, Czech Republic*
analucia.medinarodriguez@cmi.gov.cz
[2] *CTU in Prague FEE, Department of Measurements, Technická 2, 166 27 Prague 6, Czech Republic*

***Abstract* – A digital twin of a four-terminal-pair (4-TP) AC impedance bridge is being developed in LTspice, including dominant non-ideal effects and experimentally characterized parasitics. The model supports sensitivity analysis and error evaluation. An automated balancing algorithm implemented in Python shows stable convergence and sufficient numerical accuracy. Ongoing work focuses on model validation and refinement of convergence parameters.**

Keywords: Automation; complex interpolation; impedance bridge; uncertainty.

I. INTRODUCTION

The precise measurement of AC impedances is a cornerstone of electrical metrology. In recent years, reconfigurable digital impedance bridges operating in fully digital (FD) and digitally assisted (DA) modes have been developed to meet the stringent uncertainty requirements of national metrology institutes [1]. However, detailed performance analysis of physical bridges is limited by practical constraints of real instruments and often relies on time-consuming experimental investigations of multiple parameter configurations [2]. In this context, a digital replica of the bridge provides a valuable alternative, enabling systematic studies of sensitivity, error propagation, and operating conditions that are difficult to explore directly on the hardware.

II. DIGITAL TWIN AND AUTOMATED BALANCE

The study is based on a reconfigurable 4 TP FD/DA impedance bridge previously reported in [1]. A circuit level digital twin was implemented in LTspice including dominant non ideal effects, to support the development of the balancing strategy.

An automated balancing algorithm was implemented in Python using a two-point complex interpolation, where each excitation source is modeled as a complex phasor $G = Ae^{j\phi}$. To approach more realistic operation, finite generator resolution and phase discretization were included, and additive white noise was introduced at the detector readouts. Validation of specific component effects is therefore planned to use extreme-ratio measurements, as proposed in [3].

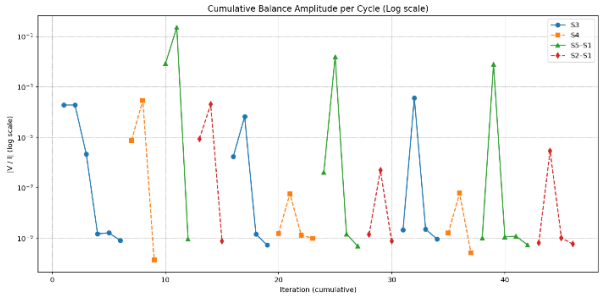


Fig. 1. Imbalance magnitude $|V|$ or $|I|$ per iteration (log scale) for each source.

III. CONCLUSION

The developed simulation reproduces the physical bridge behavior, including dominant parasitic effects. Preliminary results of the automated balancing algorithm show stable convergence comparable to the physical system. Further work focuses on implementation of realistic parameters of more involved parts, quantitative model validation and refinement of convergence parameters, and it is planned to use the modeled bridge for validation of new calculable resistors against current state-of-the-art standards.

REFERENCES

[1] J. Kučera and J. Kováč, "A reconfigurable four terminal-pair digitally assisted and fully digital impedance ratio bridge," *IEEE Trans. Instrum. Meas.*, 2018, doi: 10.1109/TIM.2018.2790538.

[2] M. Ortolano et al., "A comprehensive analysis of error sources in electronic fully-digital impedance bridges," *IEEE Trans. Instrum. Meas.*, 2020, doi: 10.1109/TIM.2020.3034115.

[3] J. Kučera, J. Kováč, P. Svoboda, and L. Bártová, "Alternative Traceability Chains of Impedance Units With Fully Digital Impedance Bridges," *IEEE Trans. Instrum. Meas.*, 2025, doi: 10.1109/TIM.2025.3552391.

The primary Tesla Standard at PTB

Franziska Weickert[1], Zhaowen Liu[1], Rainer Ketzler[1], Wiebke Pohlandt[2], Wolfgang Kilian[2]

[1]*Physikalisch-Technische Bundesanstalt, 38116 Braunschweig, franziska.weickert@ptb.de*
[2]*Physikalisch-Technische Bundesanstalt, 10587 Berlin*

***Abstract* – The presentation gives an overview on the primary Tesla standard at the German metrology institute PTB as disseminated along traceability chains to stakeholders, industry and end-users. We present long established methods and diskuss new developments.**

Currently, the primary Tesla standard is realized by nuclear magnetic resonance (NMR) techniques on protons with nuclear spin $I = \frac{1}{2}$. The Larmor frequency of the spin precession $\omega_L = \gamma B$ in a magnetic field is directly proportional to the magnetic flux density B. The proton gyromagnetic ratio γ is known with one part in 10^8 precision from particle-physics experiments[1] and it traces the unit Tesla back to the SI standard time (t). Traceable calibration services are carried out at PTB in the range of 10 μT to 0.3 T. The method of free induction decay (FID) on pure water samples is used between 10 μT and 2 mT reaching lowest measurement uncertainties (MUs) of one part in 10^6 (1 ppm) at around 1 mT.[2] At higher fields between 1 mT and 0.3 T, a NMR absorption technique[3] is utilized. Here, the NMR resonance frequency of an LC circuit is measured by means of a marginal oscillator. Samples of aqueous $CuSO_4$ solution enhance NMR absorption rates for a reasonable signal to noise ratio, however, $CuSO_4$ impurities also cause a systematic field distortion, which adds parasitic contributions to the measured magnetic field value. The accuracy of the absorption method is therefore limited to 100 ppm.

Furthermore, we discuss results on NMR measurements taken on gaseous ^{3}He samples that have an $I = \frac{1}{2}$ nuclear spin as well but exhibit significantly enhanced relaxation times compared to proton spins. Larger relaxation times combined with smaller chemical shift and less temperature dependence reduce the level of MUs for magnetic field measurements with ^{3}He samples. Additionally, hyperpolarization opens the road for a larger field range of the primary Tesla standard at both ends: above 0.3 T and below 10 μT.

REFERENCES

[1] P. J. Mohr, D. B. Newell, and B. N. Taylor,, "CODATA recommended values of the fundamental physical constants: 2014", Rev. Mod. Phys., vol.88, 2016, pp. 035009.

[2] K. Weyand, "Maintenance and Dissemination of the Magnetic Field Unit at PBT", IEEE Trans. Instr. Meas., 2001, pp. 470-473.

[3] K. Weyand, "An NMR Marginal Oscillator for Measuring Magnetic Fields Below 50 mT", IEEE Trans. Instr. Meas., vol.38, 1989, pp. 410 - 414.

A traceable calibration of low-field magnetic coil systems in noisy industrial environments

Michal Janosek[1], Michal Dressler[1], Michal Ulvr[2]

[1] *Dept. of Measurement, FEE, Czech Tech. Univ. in Prague, Technicka 2, Prague, CZ*
[2] *Laboratorie for Fundamental metrology, Czech Metrology Institute, V Botanice 4, Prague, CZ*

***Abstract* – We present a procedure for calibrating low-field (≤ 100 µT), magnetic coil systems in industrial practice where the peak magnetic field noise can reach units of percents of the range. We utilize a secondary magnetic standard, together with a fast settling linear current supply and a fast fluxgate magnetometer as a transfer standard. Using our procedure, we were able to achieve about 110 ppm uncertainty even in an environment with up to 150 nT (1500ppm) peak noise.**

Keywords: calibration, magnetic, coils, traceability.

I. INTRODUCTION

The primary standard of DC magnetic flux density at the Czech metrology institute is flowing water, "nutation" NMR teslameter [1] for fields above 0.6 mT and a Barker solenoid on quartz core for fields below 0.6 mT. The uncertainty of the NMR standard and also of the Barker solenoid constant is about 50 ppm (k=2). For an out-of-house calibration, a Garrett solenoid is used, with field constant of 2.34972 mT/A, uncertainty of 300 ppm and homogeneity ± 150 ppm in a 5-cm wide central area.

The standard calibration procedure would be placing the secondary standard in the calibrated coils and using the null-method [2]. Not only the bulkiness of the secondary coil can be a problem, but also the field zero (null) is directly affected by local noise and enters the final uncertainty. Previously, we have developed a procedure with a fast-settling current source, where fast magnetometers could be calibrated even in noisy environments [3]. We modified it by utilizing the fast fluxgate magnetometer as a short-term transfer standard.

II. CALIBRATION PROCEDURE

The fast settling magnetometer (±100 µT range) is first calibrated in the Garrett solenoid and then its constant transferred to the coil system under test. The width of fluxgate compensating coil is 1-cm, so fits in the Garett coil homogeneous area. The coil to be calibrated was a 75-cm Meritt coil (similar to [3]) in a noisy laboratory.

For coil constant calibration, we used a 6-point linear current sweep covering about 50 % of the fluxgate range (Fig. 1). Thirty of those sweeps were used to calculate the temporary fluxgate constant in the first step. Then the procedure was repeated for the Merritt coils. Currents were adjusted so that the fluxgate sees approximately the same field, to suppress any nonlinearity in its response.

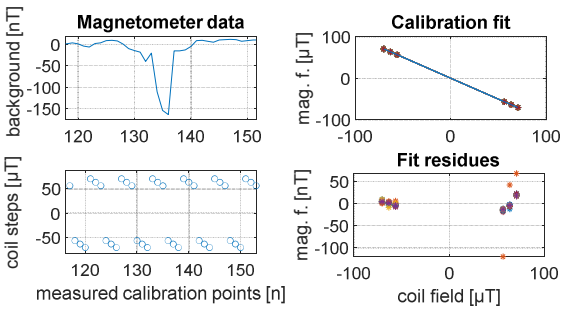


Fig. 1. The zoomed-in calibration sweeps and field noise (left) and overall calibration fit and residues (right)

III. RESULTS

The uncertainty of the Garrett-to-fluxgate constant transfer was 60 ppm (k=2, type A), and to the Meritt coil about 50 ppm (k=2, type A). If we consider misalignment in the Meritt coil (0.5 degrees: 80 ppm, k=2, type B), we end at a combined transfer uncertainty of 110 ppm (k=2). The transfer uncertainty in a noisy laboratory was comparable to what was obtained in a magnetically quiet, sub-urban location [2].

REFERENCES

[1] M. Ulvr and J. Kupec, "Improvements to the NMR method with flowing water at CMI," in IEEE Transactions on Instrumentation and Measurement, 2017, vol. 61, pp. 204-208.

[2] A. Zikmund et al., "Precise calibration method for triaxial magnetometers not requiring Earth's field compensation," in IEEE Transactions on Instrumentation and Measurement, 2015, vol. 64, pp. 1242-1247.

[3] M. Janosek, M. Dressler, V. Petrucha, and A. Chirtsov, Magnetic calibration system with interference compensation. IEEE Transactions on Magnetics, 2018, vol. 5, art. no. 6000104.

A Digital Voltage Ramp Generator with Integrating Compensation

Yaqiong Fu[1], Xianhui Li[1], Kenan Xu[1], Qing He[2], Jianting Zhao[2,*]

[1] *China Jiliang University, Hangzhou 310018, China*
[2] *National Institute of Metrology, Beijing 100029, China*
[*] *E-mail: zhaojt@nim.ac.cn*

***Abstract* – A digital voltage ramp generator with integrating compensation is proposed in this paper. The ramp for compensation is generated by a pair of controllable current sources, integrating capacitors and analog switches. Due to the fewer instructions that need to be sent in each compensation cycle, the slope rate can be further increased.**

Keywords: Digital voltage ramp generator; nonlinear compensation; software-controlled current source; analog switch.

I. INTRODUCTION

High-linearity, long-duration voltage ramps are required for picoampere current calibration, mutual inductance measurement, and CCC bridges. Existing designs use integrator-based correction [1] or dual-DAC structures [2]. This paper presents a software-controlled compensation scheme using current sources and Integrating circuit to increase slope rate while maintaining linearity.

II. CIRCUIT DESIGN

The proposed generator consists of a main circuit and a compensation circuit, as illustrated in Fig. 1. The main circuit employs a conventional DAC to generate a staircase signal. The compensation circuit comprises two identical branches; in each branch, a software-controlled current source, realized by a multiplying DAC, injects a DC current into an integrator to produce a ramp signal. The compensation slope and duration are governed by the DAC codes and timing, requiring only one instruction per cycle. The two complementary branches alternately compensate and discharge the main circuit signal. The weighted sum of the main circuit signal and the compensation signal is then output via an amplifier.

III. PERFORMANCE AND DISCUSSION

A preliminary circuit device was designed. The compensation principle and the corresponding measured results are presented in Fig. 2. It can be observed that the smoothness of the ramp signal is significantly improved. With this method, the nonlinearity of the main circuit DAC can also be compensated simultaneously. Furthermore, because only one DAC instruction is required per compensation cycle, this method enables faster generation of smooth ramp signals.

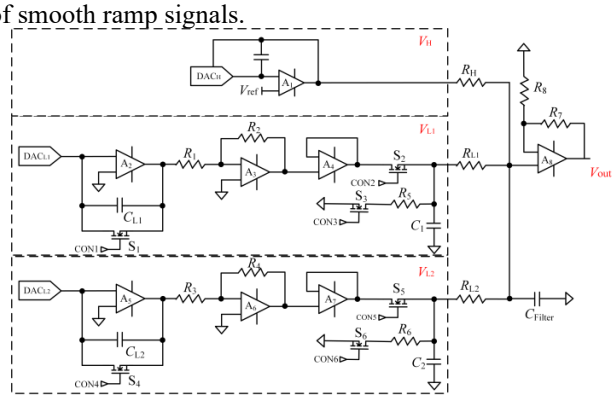


Fig. 1. Schematic diagram of the voltage ramp generator.

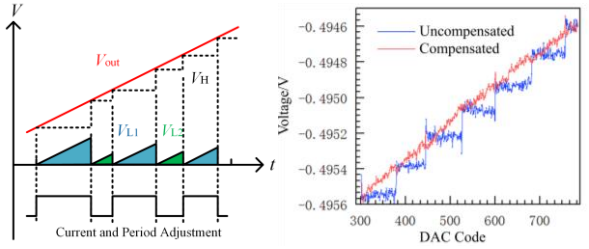


Fig. 2. Compensation principle (left figure) vs. actual compensation effect (right figure).

REFERENCES

[1] H. E. Brom, et al., "Accurate Subpicoampere Current Source Based on a Differentiating Capacitor With Software-Controlled Nonlinearity Compensation," in IEEE Transactions on Instrumentation and Measurement, vol. 54, pp. 554-558, 2005.

[2] G. D. Willenberg, et al., "Novel Digital Voltage Ramp Generator for Use in Precision Current Sources in the Picoampere Range," in IEEE Transactions on Instrumentation and Measurement, vol. 58, pp. 756-760, 2009.

Acoustic-Based Detection, Classification, and Localization of Impulsive Sounds

Jakub Svatos, Martin Maxa

Due to the increasing frequency of shootings in public spaces and the rise in global gun ownership, there is a growing demand for enhanced autonomous security in communal areas such as campuses, hospitals, parks, and government facilities. While traditional surveillance technologies like cameras, drones, and security personnel provide some level of protection, they often fall short in high-risk situations involving active shooters. An automatic acoustic surveillance system presents a valuable addition, enabling faster and more precise detection of threats while reducing response times for security forces. Acoustic-based detection systems offer significant advantages by identifying gunfire, pinpointing its source, and tracking ongoing threats. These systems leverage the distinct acoustic signatures of firearms, using sound propagation and signal processing techniques to determine the location and caliber of a weapon.

The muzzle blast and the shock wave define a gunshot's acoustic pattern. The muzzle blast results from explosive gases expanding from the gun barrel, traveling at the speed of sound in the bullet's direction and lasting 1–3 ms, often resembling an "N" shape. If the bullet speed exceeds the speed of sound ($v > c$), a secondary shock wave, or "N-wave," is produced due to compression and expansion shocks. Advanced signal processing is crucial to detect and classify gunshots accurately, as traditional time and frequency domain methods are unreliable. Spectral analysis often captures environmental noise rather than the gunshot itself, while time-domain patterns can be distorted by nonlinear dispersion and obstacles. Given these challenges, effective feature extraction must integrate time and frequency domain processing.

The proposed acoustic detection, localization, and classification system uses a distributed network of stand-alone sensors and a remote server. Each sensor continuously monitors its surroundings and transmits detected acoustic signals to a remote server for advanced processing. The system ensures broader coverage and adaptability to various environments by deploying multiple units. The system classifies acoustic events and triangulates their location using data from multiple sensors, enabling more precise and reliable 2D event localization. The system has been tested with various firearm calibers and ammunition.

For the identification of an event, an algorithm based on cepstral coefficients has been developed, see Fig. 1. Experiments revealed the effectiveness of this method as a feature extractor for further classification based on Neural Network (NN) and Support Vector Machine (SVM) classifier. The Mel-Frequency Cepstral Coefficients (MFCC) method, initially developed for other acoustic applications, has proven effective for gunshot recognition, with successful tests conducted in military shooting ranges. MFCCs are widely used for analyzing audio spectral characteristics, with modifications such as Linear-Frequency Cepstral Coefficients (LFCC) and Inverse Mel-Frequency Cepstral Coefficients (IMFCC) offering alternative filtering approaches. Other cepstral coefficients, like Linear Prediction Cepstral Coefficients (LPCC) and Perceptual Linear Prediction Coefficients (PLPC), serve various purposes, including noise elimination and speech recognition. Additionally, Greenwood Function Cepstral Coefficients (GFCC) and Gammatone-Frequency Cepstral Coefficients (GTCC), derived from human auditory response models, are used for classifying animal sounds and enhancing noise robustness in audio recognition, with GTCC offering greater resistance to noise compared to MFCC.

Experiments showed that MFCC, LFCC, and IMFCC yield similar results. At the same time, GTCC consistently performs better, as seen in Tab I. Increasing frame length improves classification accuracy but may also capture environmental reflections, potentially reducing focus on the gunshot itself. Results indicate that the NN classifier outperforms the SVM classifier due to its superior ability to model complex patterns. Overall, GTCC proves more effective than Mel-frequency cepstral coefficients, and NN achieves slightly better classification success, especially with longer frames.

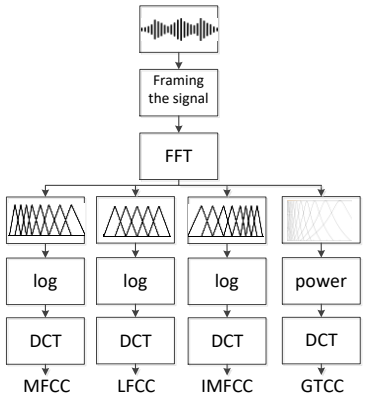


Fig. 1. The block diagram of the feature extraction algorithms.

Classifier		SVM	NN	SVM	NN	SVM	NN
Frame length	Feature extractor	ACC (%)	ACC (%)	RLC (%)	RLC (%)	MCC (-)	MCC (-)
15 ms	MFCC	83,90	85,59	80,65	67,74	0,62	0,62
	LFCC	83,90	83,90	80,65	64,52	0,62	0,57
	IMFCC	85,59	90,68	83,87	87,10	0,66	0,77
	GTCC	93,22	97,46	87,10	93,55	0,82	0,93
30 ms	MFCC	84,75	87,29	67,74	87,10	0,60	0,70
	LFCC	87,29	83,90	70,97	70,97	0,66	0,59
	IMFCC	85,59	92,37	67,74	80,65	0,62	0,80
	GTCC	97,46	98,31	90,32	93,55	0,93	0,96
50 ms	MFCC	84,75	93,22	74,19	90,32	0,61	0,83
	LFCC	88,98	86,44	77,42	77,42	0,71	0,66
	IMFCC	89,83	90,68	74,19	80,65	0,73	0,76
	GTCC	97,46	99,15	90,32	96,77	0,93	0,98

Table I. Example of Classification results for a 9 mm class.

Author: Dr, Jakub Svatos, Czech Technical University in Prague, Faculty of Electrical Engineering, Department of Measurement, Technicka 2 166 27, Prague 6, Czechia, svatoja1@fel.cvut.cz.

Lock-in detection algorithm for signals modulated with irregular intervals

Shingo Tamaru

In many electrical measurements, acquired waveforms are dominated by low-frequency noise due to drifts and various fluctuations. The lock-in detection technique, which modulates the signal of interest at a frequency higher than the noise spectrum and isolates the modulation frequency component, is widely used to recover weak signals from such noisy waveforms. However, conventional lock-in detection tacitly assumes a constant modulation interval, which is not always valid. We encountered a situation in which the signal of interest was inevitably modulated with irregular intervals during the development of transformer-coupled permeameter (TC-Permeameter) [1,2]. In such cases, the noise rejection performance of conventional lock-in algorithms severely deteriorates, resulting in poor stability and low signal-to-noise ratio of the measurement results.

This work presents a lock-in detection algorithm tailored for irregularly modulated signals [3]. Fig. 1 shows the waveforms assumed in this work. The signal being measured $\boldsymbol{s}$ is modulated with irregular intervals and an amplitude $\pm s$. The low-frequency noise $\boldsymbol{d}$ varies sufficiently slowly compared to the modulation intervals. The total waveform $\boldsymbol{v}$ is a sum of $\boldsymbol{s}$ and $\boldsymbol{d}$, i.e. $\boldsymbol{v}=\boldsymbol{s}+\boldsymbol{d}$. The measurement system is configured to acquire $\boldsymbol{v}$ synchronously with the modulation intervals. We want to evaluate the modulation amplitude s of the signal being measured $\boldsymbol{s}$ contained in the noisy total waveform $\boldsymbol{v}$.

The signal processing algorithm for lock-in detection proposed in this work is based on the generalized Savitzky-Golay differentiation filter (SGDF) formalism for irregularly sampled data. The noise waveform $\boldsymbol{d}$ is approximated by a low-order polynomial of time, and a high-order numerical differentiation vector $\boldsymbol{D}$ is derived from the time stamps of data acquisition. The application of $\boldsymbol{D}$ to $\boldsymbol{v}$ eliminates low-frequency noise $\boldsymbol{d}$ by polynomial differentiation, while preserving the signal being measured $\boldsymbol{s}$ modulated with irregular intervals.

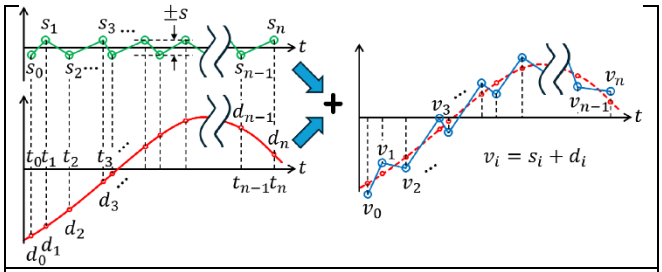


Fig. 1, Signal waveforms assumed in this work.

Fig. 2 shows an example of raw data obtained by TC-Permeameter. It alternatively changes the bias magnetic field H_B between two values and acquires the transmission parameter S_{21} under the two H_B, which is shown as red and blue circles, respectively. The difference of S_{21} (vertical axis) between red and blue circles is converted to the permeability of the sample, which thus corresponds to the modulation amplitude s that needs to be evaluated in this measurement. However, the raw data contain large drifts and fluctuations, and the modulation intervals are not constant but have large offset.

This measurement was repeated 10 times, and both the conventional and the proposed algorithms were applied to the same datasets for data analysis. As shown in Fig. 3, the standard deviation (SD) of the results analysed by the proposed algorithm is nearly two orders of magnitude smaller than that by the conventional one. This comparison clearly demonstrates that the algorithm proposed in this work efficiently rejects low-frequency noises even when the modulation interval is irregular, while the conventional one fails to do so. Further details, including mathematical formalism and broader applicability of the algorithm, will be discussed in the presentation.

Acknowledgements: This work is supported in part by the MIC/SCOPE Grant Number JP195003002, and in part by MEXT-Program for Creation of Innovative Core Technology for Power Electronics (INNOPEL) Grant Number JPJ009777.

References: [1] J. Magn. Magn. Mater., vol. 501, p. 166434 (2020). [2] Rev. Sci. Instrum., vol. 94, p. 073901 (2023). [3] Measurement, p 117894 (2025)

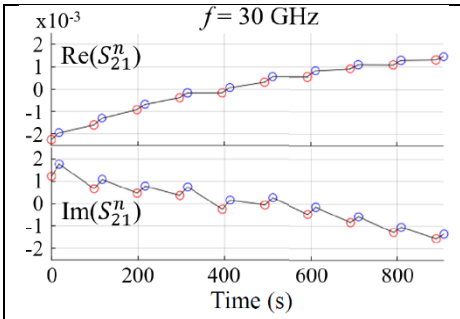


Fig. 2, An example of raw data obtained by TC-Permeameter.

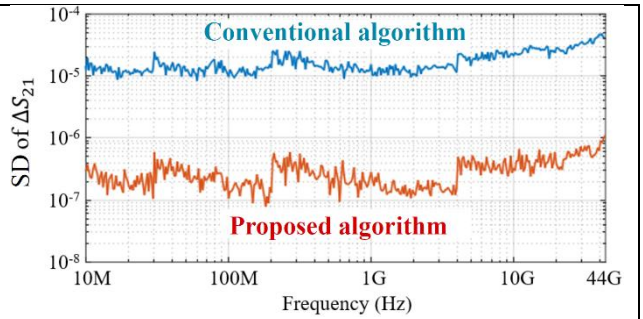


Fig. 3, Comparison of SD of the 10 measurement results analysed by two lock-in detection algorithms.

Author: Dr. Shingo Tamaru, Research Institute for Hybrid Functional Integration (HyFI-RI), National Institute of Advanced Industrial Science and Technology (AIST), Central 2, 1-1-1 Umezono, Tsukuba, Ibaraki, 305-8568, Japan, +81-50-3521-3206, shingo.tamaru@aist.go.jp

A Wireless Acoustic Monitoring System for Drone Localization Based on MEMS Microphones and LoRa Communication

Kamil Herman[1], Jakub Svatos[1], Martin Maxa[1]

[1]*Czech Technical University in Prague, Faculty of Electrical Engineering, Department of Measurement, Technicka 2, 160 00 Prague 6, Czech Republic, hermakam@fel.cvut.cz*

***Abstract* – This paper presents the design and initial development of a distributed wireless acoustic sensor network for drone detection and localization. Based on a star topology, multiple autonomous measuring nodes communicate with a central hub via LoRa technology. Each node is equipped with an STM32H5 MCU and Infineon IM67D130AXTSA2 MEMS microphones. GPS is integrated to determine the unit's geographical coordinates and verify field placement. To ensure robustness in environments with limited GPS availability, an IMU is included for auxiliary localization support. The hardware is encased in a UV-resistant ASA 3D-printed housing. The paper describes the hardware architecture and discusses the potential of the system for low-power long-term monitoring.**

Keywords: Drone detection; Acoustic monitoring; LoRa; MEMS microphones.

PROPOSED ACOUSTIC MONITORING SYSTEM

A. *Introduction*

The proliferation of UAVs poses significant security challenges [1], [2]. While radar systems are common, they often exhibit a "blind zone" or reduced sensitivity in short ranges. Acoustic detection offers a cost-effective, passive alternative that addresses this gap. Unlike visual methods, acoustic monitoring does not require direct line-of-sight, making it effective in obstructed environments. Various techniques have been explored [3], [4]. This work focus on a distributed measurement system for surveillance using low-power acoustic sensors.

B. *System Architecture*

The system utilizes a star topology for centralized data aggregation and management. The general principle of the distributed network and interaction between components is illustrated in Fig. 1.

C. *Implementation Details*

Autonomous nodes are built around STM32H5 MCUs and Infineon IM67D130AXTSA2 MEMS microphones. The units include a u-blox NEO-M8 GPS for coordinates and an IMU to maintain localization in environments with limited GPS. For logging, an SD card slot is provided. Power is supplied by a high-capacity Li-Pol battery, protected by a UV-resistant ASA 3D-printed case. The central hub acts as a gateway utilizing LoRa for efficient long-range communication and Wi-Fi for local data access.

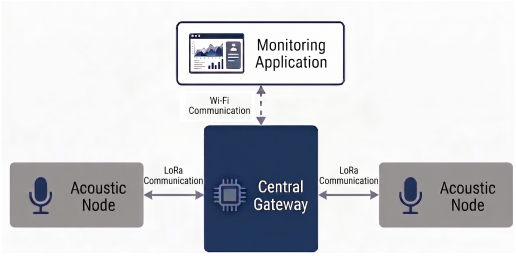


Fig. 1. Principle schema of the monitoring system.

D. *Detection Principle*

Detection is based on spectral analysis of acoustic signatures produced by drone motors. Each node performs initial signal processing to identify narrow-band components and harmonics. Localization is performed in the central gateway by analyzing the acoustic signals from multiple nodes, leveraging signal amplitude or timing information across the star network topology.

E. *Conclusion*

The architecture offers a scalable and robust solution for acoustic drone monitoring. By utilizing distributed low-power nodes based on STM32H5 and LoRa, the system enables surveillance with minimal infrastructure. The integration of MEMS microphones and IMU ensures reliable operation in challenging environments. This design provides a solid foundation for passive acoustic surveillance networks.

REFERENCES

[1] M. A. Khan, et al., "On the Detection of Unauthorized Drones", IEEE Sensors J., 22(12), 2022, ISSN 1530-437X.
[2] I. Guvenc, et al., "Detection and Interdiction for Amateur Drones", IEEE Comm. Mag., 56(4), 2018, ISSN 0163-6804.
[3] P. Nguyen, et al., "Investigating RF-based Detection of Drones", Proc. DroNet '16, 2016, ISBN 9781450344050.
[4] P. Zhu, et al., "Detection and Tracking Meet Drones Challenge", IEEE TPAMI, 44(11), 2022, ISSN 0162-8828.

Non-Destructive Microwave Dielectric Characterization of Leather for Automotive Applications

Andrea Alimenti[1], Kostiantyn Torokhtii[1], Alessandro Magalotti[1], Serena De Santis[1], Veronica Granata[1], Nicola Pompeo[1], Vincenzo Roberti[2], Daniele Rocco[1], Enrico Silva[1]

[1] *Department of Industrial, Electronic and Mechanical Engineering (DIIEM), Roma Tre University, Rome, Italy*
[2] *TME S.r.l., Caserta, Italy*
kostiantyn.torokhtii@uniroma3.it

I. ABSTRACT

The automotive industry increasingly requires advanced non-destructive methods [1] for assessing the quality and conservation state of high-value leather used in premium interiors. Since leather is a hygroscopic collagen-based material, its mechanical properties, dimensional stability, surface appearance, and long-term durability are strongly influenced by environmental humidity and absorbed moisture [2]. In this work, a microwave resonant technique is proposed for the dielectric characterization of natural bovine leather under controlled humidity conditions. The method is based on a high-Q dielectric-loaded resonator [3] operating near 10 GHz, specifically designed for an enhanced sensitivity to the complex permittivity of thin and lossy materials.

Two leather samples with different thicknesses (1.6 mm and 0.9 mm) were conditioned for 48 h in sealed environments at four relative humidity (RH) levels, ranging from dry conditions (RH < 10%) up to 80% RH. From the measurements of the resonance frequency and quality factor of the used dielectric loaded resonator the dielectric constant was extracted by means of finite-element calibration curves.

Experimental results show a clear and repeatable decrease of the resonant frequency with increasing humidity, corresponding to an increase in the real part of the relative permittivity. For both samples, ε' increased from approximately 2 in dry conditions to values around 3.2 at high humidity levels. The thicker and more homogeneous sample exhibited the best linearity and repeatability, while the thinner specimen showed larger variability due to reduced electromagnetic perturbation and surface irregularities. Standard uncertainty on the extracted permittivity remained below 3%.

The results demonstrate that microwave dielectric-loaded resonators provide a sensitive, contactless, and fully non-destructive tool for monitoring humidity-related changes in leather. The proposed approach is therefore promising for future inline quality-control systems and smart handling platforms for automotive leather manufacturing, where rapid and reliable evaluation of material state is required.

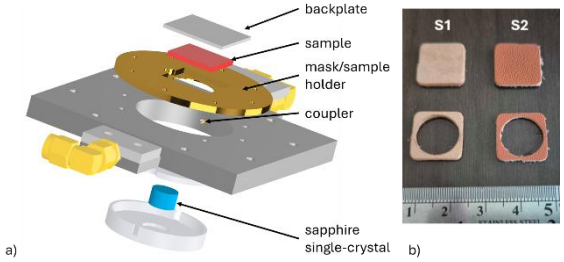


Fig. 1 Representation of the measurement cell (a) with a photo of samples under study. (b) Photograph of the measured samples S1 and S2. The top samples are those for which the dielectric permittivity was measured, while the bottom samples were used as references.

REFERENCES

[1] A. I. Hernandez-Serrano, S. C. Corzo-Garcia, E. Garcia-Sanchez, M. Alfaro, and E. Castro-Camus, "Quality control of leather by terahertz timedomain spectroscopy," Appl. Opt., vol. 53, no. 33, pp. 7872–7876, Nov. 2014.

[2] C. E. Weir, "Influence of temperature and moisture on the electrical properties of leather," Journal of research of the National Bureau of Standards, vol. 48, p. 349, 1952.

[3] A. Alimenti, E. Pittella, K. Torokhtii, N. Pompeo, E. Piuzzi, and E. Silva, "A dielectric loaded resonator for the measurement of the complex permittivity of dielectric substrates," IEEE Transactions on Instrumentation and Measurement, vol. 72, pp. 1–9, 2023.

Measuring the Future
27th IMEKO TC4 Conference 2026 with Special TC11 Technical Session
Book of Abstracts

Editors

Assoc. Prof. Jakub Svatos, Ph.D., Czech Technical University in Prague, Czech Republic (*svatoja1@fel.cvut.cz*)

Ing. Kamil Herman, Czech Technical University in Prague, Czech Republic (*hermakam@fel.cvut.cz*)

Czech Technical University in Prague

Jugoslávských partyzánů 1580/3
160 00 Prague 6 - Dejvice
Czech Republic

© 2026 CTU in Prague

All rights reserved. No part of this publication may be reproduced in any form, nor may it be stored in a retrieval system or transmitted in any form, without written permission from the copyright holders.

The abstracts are published as submitted by the authors, who are solely responsible for their content.